

the end of training:

$$\underset{p_1, \dots, p_T \in \Delta^{k-1}}{\text{minimize}} \quad \frac{1}{m} \sum_{j=1}^m L_{\text{eval},j}(f_T). \quad (1)$$

This optimization problem is challenging to solve without additional assumptions. In order to make the problem tractable, we impose an explicit dynamics rule for the each evaluation skill’s loss $L_{\text{eval},j}$ in terms of the current loss and data mixture. Assuming for simplicity that $\mathcal{S}_{\text{eval}} \subseteq \mathcal{S}_{\text{train}}$, a simple rule would be $L_{\text{eval},j}(f_t) = L_{\text{eval},j}(\Phi(f_{t-1}, p_{t-1})) := L_{\text{eval},j}(f_{t-1})(1 - \alpha p_{t-1}^j)$ for $\alpha \in [0, 1]$. That is, we expect that allocating more data to skill j should result in the validation loss on skill j decreasing. However, such an expression assumes that only training on the j th skill will help learn the j th skill. Instead, Section 2.2 suggests that there are other skills that may help with the j th skill. We propose the following dynamics:

$$L_{\text{eval},j}(f_t) = L_{\text{eval},j}(f_{t-1})(1 - A_{:,j}^\top p_{t-1}), \quad (2)$$

where $A_{:,j}$ is the column with weights of all skills that influence $s_{\text{eval},j}$, and we absorb the scalar α into A . The optimization problem in (1) can thus be simplified as follows:

$$\begin{aligned} & \underset{p_1, \dots, p_T \in \Delta^{k-1}}{\text{minimize}} \quad \frac{1}{m} \sum_{j=1}^m L_{\text{eval},j}(f_T) \\ & \text{s.t.} \quad f_t = \Phi(f_{t-1}, p_{t-1}) \quad \forall t = 1, \dots, T \\ & \quad \quad L_{\text{eval},j}(f_t) = L_{\text{eval},j}(f_{t-1})(1 - A_{:,j}^\top p_{t-1}) \quad \forall j \in [m] \end{aligned} \quad (3)$$

In Appendix B, we derive the following update rule via online mirror descent [45] for learning rate $\eta > 0$:

$$p_{t+1}^i = p_t^i \exp \left(\eta \sum_{j=1}^m A_{ij} L_{\text{eval},j}(f_t) \right). \quad (4)$$

In addition, when equation 4 is expanded, we have that $p_{t+1}^i = p_1^i \exp \left(\eta \sum_{\tau=1}^t \sum_{j=1}^m A_{ij} L_{\text{eval},j}(f_\tau) \right)$. Since this summation over τ results in diminishing strength of updates, we change it to a moving window of size w . Our full method is in Algorithm 1.

Intuitively, at each step we adjust the weight on skill i based on the losses of skills that i influences, with the assumption that more training data helps decrease loss. Note that when we use our algorithm with a complete graph or empty graph, we achieve expected behavior discussed in Section 2.1. For the complete graph, our algorithm reduces to stratified sampling. When we have a skill set with an empty graph, the update rule reduces to sampling proportional to each skill’s validation loss.

4 Experimental results

Given an ordered skill set, we aim to validate SKILL-IT’s ability to select data for efficiently learning skills in the continual pre-training, fine-tuning, and out-of-domain settings. We provide full tables of results in Appendix D.3.1 and results where we learn the skills graph on the 125M model and use it for the 1.3B parameter model in Appendix D.4. Skills graphs are in Appendix C.2, weight trajectories for SKILL-IT are in Appendix D.3.2, and ablations on the graph and online components of SKILL-IT are in Appendix D.5.

4.1 Continual pre-training

Setup We evaluate the ability of SKILL-IT to select data for efficiently learning over all skills. We measure average validation loss per skill after a fixed number of training steps. We construct the LEGO synthetic and addition synthetic with $k = 5$ and 3, respectively, and an imbalanced dataset over the skills. On the Natural Instructions dataset, we use 23 of the task categories as skills.

Baselines We compare SKILL-IT against three baselines that do not account for skills: random sampling, curriculum learning, and anticurriculum learning. Random sampling is a standard procedure for selecting samples given no additional information. Curriculum learning [3] and anticurriculum learning [67] score the samples from easiest to hardest and vice versa, respectively, and sample over an expanding set of the lowest scored samples at every epoch; we use the pre-trained

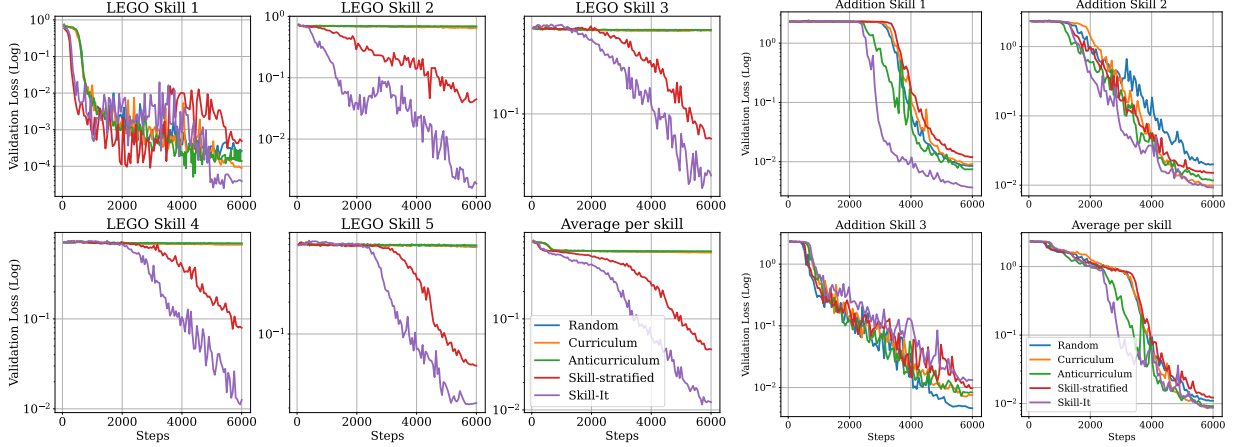


Figure 4: Performance of SKILL-IT on each skill in the continual pre-training setting (learning over all skills in the ordered training skill set) on the LEGO synthetic (left) and addition synthetic (right).

model’s loss to rank points. We evaluate skill-stratified sampling, which uses knowledge of the skills but is not online, and include an additional skills curriculum baseline in Appendix D.3.1

Analysis Our results are shown in Figure 4. Across our experiments we find that SKILL-IT outperforms baselines that do not use skills as well as skill-stratified sampling. On the LEGO dataset, all three baselines that do not utilize a notion of skills exhibit plateauing loss on four of the skills. Both skill-stratified sampling and SKILL-IT are able to significantly reduce loss on all skills, but the former is slower. Halfway through training, SKILL-IT exhibits an accuracy improvement between 9.9 and 25.9 points over other approaches, reaching a final accuracy of 99.4 (Figure 19). SKILL-IT outperforms skill-stratified sampling by initially allocating more weight to prerequisite skills and eventually allocating more weight to skills that are learned more slowly (Figure 20). On the addition synthetic with $k = 3$, SKILL-IT converges to near-zero validation loss faster than the baselines on skills 1 and 2. While the random baseline may seem competitive at first glance, it fails to learn skill 1 (adding together the ones digits), which hurts its average loss per skill. On NI, the validation loss from SKILL-IT is 3.2% lower than from random sampling (Table 7). Our results suggest that exploiting the construction and ordering of skills is critical to learning skills quickly.

4.2 Fine-tuning

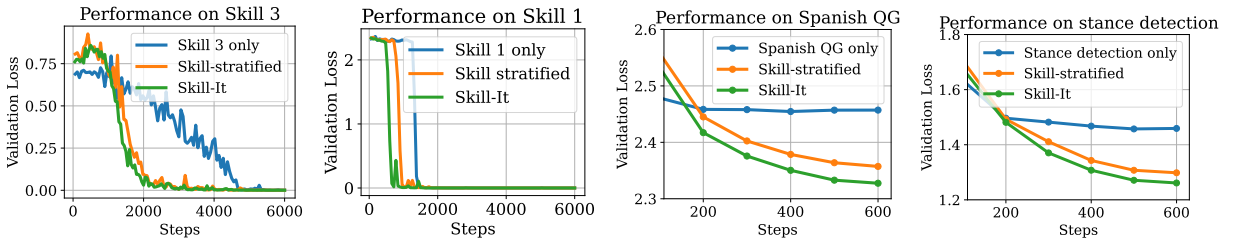


Figure 5: Performance of SKILL-IT in the fine-tuning setting (learning a target skill using the ordered training skill set) on LEGO, addition, and NI.

Setup We evaluate the ability of SKILL-IT to select data from an ordered training skill set for learning a target skill. Mirroring Figure 3, we evaluate on LEGO target skill 3 (third in reasoning chain), on the addition synthetic’s skill 1 (ones place digit addition), and on NI’s Spanish QG and Stance Detection.

Baselines We compare SKILL-IT against training on the target skill only and skill-stratified sampling over prerequisite skills and the target skill. The skill-stratified sampling approach uses the ordered skill set to identify prerequisite skills, but does not exploit them dynamically.

Analysis Our results are shown in Figure 5. On LEGO, SKILL-IT results in the same validation loss of 0.01 as training only on the target skill in 38.1% fewer steps. We observe a similar trend on addition, with SKILL-IT converging to a validation loss of 0.01 in 59% fewer steps required to do so when training only on the target skill. Finally, on NI, SKILL-IT improves validation loss on Spanish question generation by 5.3% and Stance Detection by 13.6% over just training on the respective

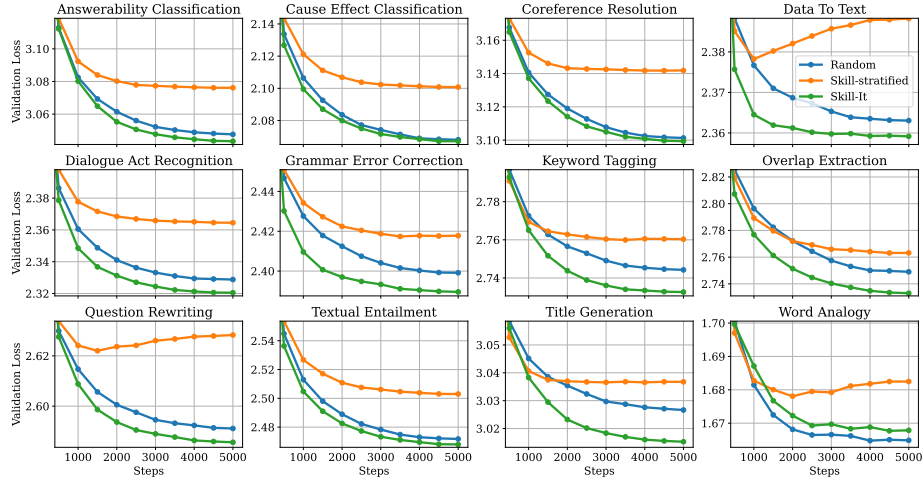
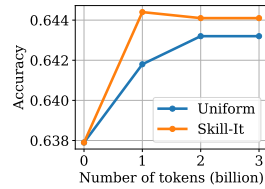


Figure 6: Performance of SKILL-IT in the out-of-domain setting for the NI test task split. SKILL-IT uses the graph between the train and evaluation skills to produce an online mixture on the training dataset.



RedPajama source	SKILL-IT mixture
ArXiv	0.1370
Books	0.0437
C4	0.4195
CommonCrawl	0.0732
GitHub	0.189
StackExchange	0.0892
Wikipedia	0.0484

Figure 7: Left: Accuracy on LM Evaluation Harness for continual pre-training of a 3B parameter model using SKILL-IT on the RedPajama dataset. We achieve higher accuracy at 1B additional tokens than uniform at 3B tokens. Right: SKILL-IT mixture over RedPajama sources.

target skill only. In this setting, a significant portion of the improvement over training only on the target skill comes from identification of prerequisite skills through the learned graph in the skill-stratified sampling method. SKILL-IT is further able to improve performance with finer-grained dynamic weighting on prerequisite skills.

4.3 Out-of-domain setting

Natural Instructions We evaluate the ability of SKILL-IT to select data from a set of training skills for learning a disjoint set of evaluation skills that we cannot train on. We use all 59 task categories in the NI train tasks split as the training skills and the 12 task categories in the test tasks split as our evaluation skills. We compare SKILL-IT against random and skill-stratified sampling, both of which do not exploit the relationships between training skills and evaluation skills. SKILL-IT achieves the lowest loss on 11 out of 12 task categories over random and skill-stratified sampling (Figure 6, tables in Appendix).

RedPajama We use SKILL-IT to produce a data mixture on the RedPajama dataset. The training skills are the data sources comprising the dataset, and the evaluation skills are several tasks from the Language Model Evaluation Harness [14]. SKILL-IT with $T = 1$ (i.e. a static, graph-based mixture) yields the mixture in Figure 7 (right). We continually pre-train a 3B parameter model trained on one trillion tokens for three billion additional tokens using this mixture, and see that it outperforms uniform sampling over the data sources (Figure 7 left). In particular, SKILL-IT achieves higher accuracy with 1B additional tokens than uniform with 3B additional tokens.

5 Related work

Data selection for LMs There have been several studies of large-scale data selection for LMs. Data deduplication [1, 22, 32], in which identical or nearly identical samples are removed, is a method that enables LMs to be trained on smaller, cleaned datasets and has been increasingly used as a pre-processing step for training data [4, 59, 71]. Other methods applied at scale involve ensuring high quality of data by explicitly filtering out samples or comparing the training dataset with a cleaned reference dataset [7, 31, 59]. Importance reweighting approaches have also been proposed for identifying training data from a large corpus that best approximates a smaller target distribution [69], and influence functions have been used to select a subset of training data to improve performance on downstream tasks [61]. These approaches can identify data